

Problem Set 2 – Set identities

Session 2 · Sets & set operations

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution – say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

TWO FORMATS, AND BOTH ARE REQUIRED EXACTLY To prove an identity: double inclusion. Take an arbitrary element of one side, show it lies in the other, then do the reverse. The word "arbitrary" must appear.
To disprove one: name your sets explicitly, compute both sides, and point at the element that is in one and not the other. A Venn diagram is not a proof and is not a disproof.

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

2 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- a. $1 \in C$ (b) $\{1\} \in C$ (c) $\{1\} \subseteq C$ (d) $\{1, 2\} \subseteq C$ (e) $\emptyset \subseteq C$ (f) $|C| = 4$

3 POWER SET

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for any finite A .

Part B · Six identities to prove

FOR EVERY PROBLEM IN THIS PART Prove by double inclusion – both directions are required. A proof that verifies the identity on one example is **not a proof**, however carefully the example is computed.

4 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

5 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

6 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

7 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

8 SUBSETS

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

9 CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

Part C · Three to disprove, and one to write

10 DISPROOF

Each statement below is **false**. For each, give explicit sets, compute both sides, and name an element that is in one side and not the other.

- $(A \cup B) \setminus B = A$ for all sets A, B .
- $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for all sets A, B .
- $A \times B = B \times A$ for all sets A, B .

11 FIND THE ERROR

A student submits the following. Identify the first step that is wrong, say why it is wrong, and state whether the claimed identity is nevertheless true.

Claim. $A \setminus (B \setminus C) = (A \setminus B) \cup C$ for all sets A, B, C .

"Proof." Let $x \in A \setminus (B \setminus C)$. Then $x \in A$ and $x \notin B \setminus C$. Now $x \notin B \setminus C$ means $x \notin B$ or $x \in C$. So $x \in A \setminus B$ or $x \in C$, hence $x \in (A \setminus B) \cup C$. Therefore the two sets are equal. ■

In no more than five sentences: a Venn diagram convinced you that some identity was true, and then you were asked to prove it by double inclusion anyway. What does the proof give you that the picture does not? You are not being asked to prove anything here.
